LAPORAN WORKSHOP

SISTEM INFORMASI MOBILE FRAMEWORK

MINGGU 11

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DAFTAR ISI

DAFTAR ISI ii
DAFTAR GAMBAR iii
BAB I PEMBAHASAN 1
A. Tugas 1
B. Penyelesaian 1
1. Tampilan input page. 1
2. Tampilan output page. 1
3. Source main.dart. 2

DAFTAR GAMBAR

Gambar 1. 1 Tampilan input page 1
Gambar 1. 2 Tampilan output page 1
Gambar 1. 3 source main.dart 2
Gambar 1. 4 source data_diri_controller.dart 2
Gambar 1. 5 source data_diri_controller.dart 3
Gambar 1. 6 source data_diri_controller.dart 3
Gambar 1. 7 source input_page.dart 4
Gambar 1. 8 source input_page.dart 4
Gambar 1. 9 source input_page.dart 5
Gambar 1. 10 source input_page.dart 5
Gambar 1. 11 source input_page.dart 6
Gambar 1. 12 source output_page.dart 6
Gambar 1. 13 source output_page.dart 7
Gambar 1. 14 source output_page.dart 7
Gambar 1. 15 source output_page.dart 8
Gambar 1. 16 source output_page.dart 8
Gambar 1. 17 source output_page.dart 9
Gambar 1. 18 source output_page.dart 9
Gambar 1. 19 source output_page.dart 10

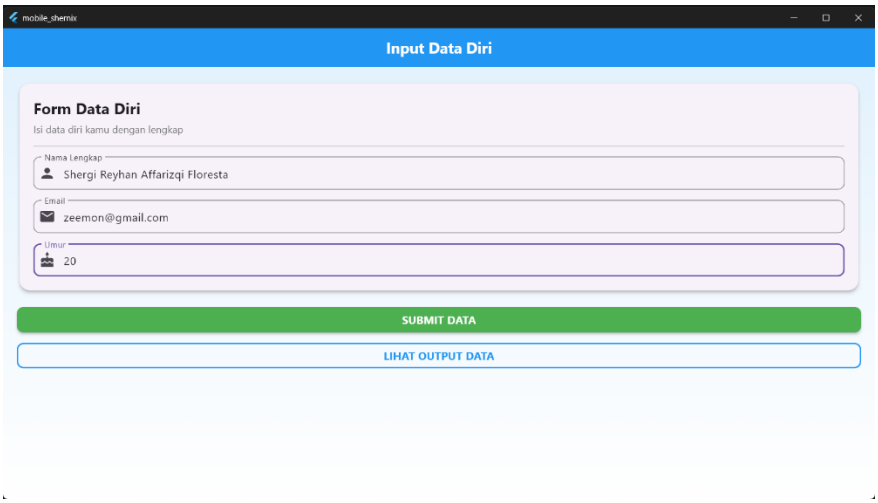
BAB I PEMBAHASAN

A. Tugas

Praktikum implementasi GetX interval acara 11.

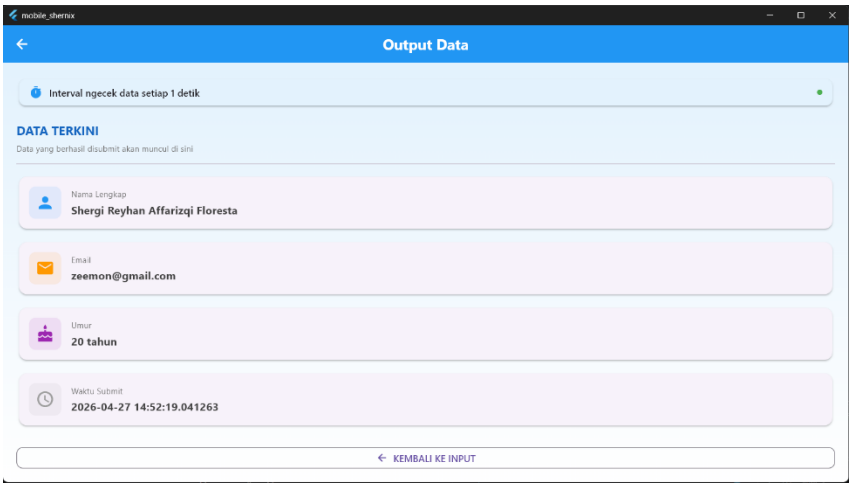
B. Penyelesaian

1. Tampilan input page.



Gambar 1. 1 Tampilan input page

2. Tampilan output page.



Gambar 1. 2 Tampilan output page

3. Source main.dart.

```dart
import 'package:flutter/material.dart';
import 'package:get/get.dart';
import 'pages/input_page.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return GetMaterialApp(
      title: 'GetX Interval Demo',
      theme: ThemeData(primarySwatch: Colors.blue, useMaterial3: true),
      home: InputPage(),
      debugShowCheckedModeBanner: false,
    ); // GetMaterialApp
  }
}
```

Gambar 1. 3 source main.dart

4. Source data_diri_controller.dart.

```dart
class DataDiriController extends GetxController {

  // Method gawe ngecek perubahan data
  void checkForDataChanges() {
    // Buat string representasi dari data saat ini
    String currentData =
        "${submittedData['nama']}|${submittedData['email']}|${submittedData['umur']}";

    // Bandingin dengan data sebelumnya
    if (currentData != lastCheckValue.value && submittedData.isNotEmpty) {
      //lak onok perubahan data
      isDataChanged.value = true;
      print("PERUBAHAN DATA DETECTED!");
      print("   Nama: ${submittedData['nama']}");
      print("   Email: ${submittedData['email']}");
      print("   Umur: ${submittedData['umur']}");
      lastCheckValue.value = currentData;

      // Reset flag setelah 2 detik
      Future.delayed(Duration(seconds: 2), () {
        isDataChanged.value = false;
      }); // Future.delayed
    }
    // else if (submittedData.isNotEmpty) {
    //   print("Tidak ada perubahan data, interval tetep jalan...");
    // }
  }

  // Method gawe submit data dari input page
  void submitData() {
    if (nama.value.isEmpty || email.value.isEmpty) {
      Get.snackbar(
        "Error",
        "Nama dan email harus diisi bro!",
        snackPosition: SnackPosition.BOTTOM,
        backgroundColor: Colors.red,
        colorText: Colors.white,
```

Gambar 1. 4 source data_diri_controller.dart

```dart
class DataDiriController extends GetxController {
  // Method gawe submit data dari input page
  void submitData() {
    if (nama.value.isEmpty || email.value.isEmpty) {
      Get.snackbar(
        "Error",
        "Nama dan email harus diisi bro!",
        snackPosition: SnackPosition.BOTTOM,
        backgroundColor: Colors.red,
        colorText: Colors.white,
      );
      return;
    }

    print("📝 User submit data: ${nama.value}");     Info: Don't invoke 'print' in production code.

    // Update data seng disubmit
    submittedData.value = {
      'nama': nama.value,
      'email': email.value,
      'umur': umur.value.isEmpty ? "-" : umur.value,
      'waktu_submit': DateTime.now().toString(),
    };

    // Reset input form
    nama.value = "";
    email.value = "";
    umur.value = "";

    Get.snackbar(
      "Sukses",
      "Data disimpan! Interval akan mendeteksi perubahan dalam 1 detik",
      snackPosition: SnackPosition.BOTTOM,
      backgroundColor: Colors.green,
      colorText: Colors.white,
    );
  }
```

Gambar 1. 5 source data_diri_controller.dart

```dart
class DataDiriController extends GetxController {

  @override
  void onClose() {
    print("Controller ditutup, interval didispose");     Info: Don't invoke 'print' in product
    intervalWorker.dispose();
    super.onClose();
  }
}
```

Gambar 1. 6 source data_diri_controller.dart

5. Source input_page.dart.

```
main.dart M   data_diri_controller.dart   input_page.dart   output_page.dart   widget_test   BLACKBOX
lib > pages > input_page.dart > InputPage > build
1  import 'package:flutter/material.dart';
2  import 'package:get/get.dart';
3  import '../controllers/data_diri_controller.dart';
4  import 'output_page.dart';
5
6  class InputPage extends StatelessWidget {      Info: Constructors for public widgets should hav
7    final DataDiriController controller = Get.put(DataDiriController());
8
9    @override
10   Widget build(BuildContext context) {
11     return Scaffold(
12       appBar: AppBar(
13         title: Text(
14           "Input Data Diri",
15           style: TextStyle(fontWeight: FontWeight.bold),
16         ), // Text
17         backgroundColor: Colors.blue,
18         foregroundColor: Colors.white,
19         centerTitle: true,
20         elevation: 0,
21       ), // AppBar
22       body: Container(
23         decoration: BoxDecoration(
24           gradient: LinearGradient(
25             begin: Alignment.topCenter,
26             end: Alignment.bottomCenter,
27             colors: [Colors.blue.shade50, Colors.white],
28           ), // LinearGradient
29         ), // BoxDecoration
30         child: Padding(
31           padding: EdgeInsets.all(20),
32           child: Column(
33             crossAxisAlignment: CrossAxisAlignment.start,
34             children: [
35               Card(
36                 elevation: 4,
37                 shape: RoundedRectangleBorder(
```

Gambar 1. 7 source input_page.dart

```
main.dart M   data_diri_controller.dart   input_page.dart   output_page.dart   widget_test   BLACKBOX
lib > pages > input_page.dart > InputPage > build
6  class InputPage extends StatelessWidget {      Info: Constructors for public widgets should hav
10   Widget build(BuildContext context) {
37                 shape: RoundedRectangleBorder(
38                   borderRadius: BorderRadius.circular(15),
39                 ), // RoundedRectangleBorder
40                 child: Padding(
41                   padding: EdgeInsets.all(20),
42                   child: Column(
43                     crossAxisAlignment: CrossAxisAlignment.start,
44                     children: [
45                       Text(
46                         "Form Data Diri",
47                         style: TextStyle(
48                           fontSize: 22,
49                           fontWeight: FontWeight.bold,
50                         ), // TextStyle
51                       ), // Text
52                       SizedBox(height: 5),
53                       Text(
54                         "Isi data diri kamu dengan lengkap",
55                         style: TextStyle(color: Colors.grey[600]),
56                       ), // Text
57                       Divider(height: 30),
58
59                       // Input Nama
60                       TextField(
61                         onChanged: (value) => controller.nama.value = value,
62                         decoration: InputDecoration(
63                           labelText: "Nama Lengkap",
64                           hintText: "Contoh: Muhammad Rizki",
65                           border: OutlineInputBorder(
66                             borderRadius: BorderRadius.circular(10),
67                           ), // OutlineInputBorder
68                           prefixIcon: Icon(Icons.person),
69                         ), // InputDecoration
70                       ), // TextField
71                       SizedBox(height: 15),
```

Gambar 1. 8 source input_page.dart

```dart
class InputPage extends StatelessWidget {
  Widget build(BuildContext context) {
                // Input Email
                TextField(
                  onChanged: (value) => controller.email.value = value,
                  decoration: InputDecoration(
                    labelText: "Email",
                    hintText: "contoh@email.com",
                    border: OutlineInputBorder(
                      borderRadius: BorderRadius.circular(10),
                    ), // OutlineInputBorder
                    prefixIcon: Icon(Icons.email),
                  ), // InputDecoration
                  keyboardType: TextInputType.emailAddress,
                ), // TextField
                SizedBox(height: 15),

                // Input Umur
                TextField(
                  onChanged: (value) => controller.umur.value = value,
                  decoration: InputDecoration(
                    labelText: "Umur",
                    hintText: "Contoh: 25",
                    border: OutlineInputBorder(
                      borderRadius: BorderRadius.circular(10),
                    ), // OutlineInputBorder
                    prefixIcon: Icon(Icons.cake),
                  ), // InputDecoration
                  keyboardType: TextInputType.number,
                ), // TextField
              ],
            ), // Column
          ), // Padding
        ), // Card

        SizedBox(height: 20),
```

Gambar 1. 9 source input_page.dart

```dart
class InputPage extends StatelessWidget {
  Widget build(BuildContext context) {
        // Tombol Submit
        SizedBox(
          width: double.infinity,
          child: ElevatedButton(
            onPressed: () => controller.submitData(),
            style: ElevatedButton.styleFrom(
              backgroundColor: Colors.green,
              foregroundColor: Colors.white,
              padding: EdgeInsets.symmetric(vertical: 15),
              shape: RoundedRectangleBorder(
                borderRadius: BorderRadius.circular(10),
              ), // RoundedRectangleBorder
              elevation: 3,
            ),
            child: Text(
              "SUBMIT DATA",
              style: TextStyle(fontSize: 16, fontWeight: FontWeight.bold),
            ), // Text
          ), // ElevatedButton
        ), // SizedBox

        SizedBox(height: 15),

        // Tombol ke halaman output
        SizedBox(
          width: double.infinity,
          child: OutlinedButton(
            onPressed: () => Get.to(() => OutputPage()),
            style: OutlinedButton.styleFrom(
              foregroundColor: Colors.blue,
              padding: EdgeInsets.symmetric(vertical: 15),
              shape: RoundedRectangleBorder(
                borderRadius: BorderRadius.circular(10),
              ), // RoundedRectangleBorder
              side: BorderSide(color: Colors.blue, width: 2),
```

Gambar 1. 10 source input_page.dart

```
main.dart M   data_diri_controller.dart ●   input_page.dart ×   output_page.dart   widget_test   BLACKBOX
lib > pages > input_page.dart > InputPage > build
6   class InputPage extends StatelessWidget {      Info: Constructors for public widgets should hav
10    Widget build(BuildContext context) {
142                 side: BorderSide(color: Colors.blue, width: 2),
143               ),
144               child: Text(
145                 "LIHAT OUTPUT DATA",
146                 style: TextStyle(fontSize: 16, fontWeight: FontWeight.bold),
147               ), // Text
148             ), // OutlinedButton
149           ), // SizedBox
150         ],
151       ), // Column
152     ), // Padding
153   ), // Container
154 ); // Scaffold
155   }
156 }
157
```

Gambar 1. 11 source input_page.dart

6. Source output_page.dart.

```
main.dart M   data_diri_controller.dart ●   input_page.dart   output_page.dart ×   widget_test   BLACKBOX
lib > pages > output_page.dart > ...
1   import 'package:flutter/material.dart';
2   import 'package:get/get.dart';
3   import '../controllers/data_diri_controller.dart';
4
5   class OutputPage extends StatelessWidget {      Info: Constructors for public widgets should ha
6     final DataDiriController controller = Get.find();
7
8     @override
9     Widget build(BuildContext context) {
10      return Scaffold(
11        appBar: AppBar(
12          title: Text(
13            "Output Data",
14            style: TextStyle(fontWeight: FontWeight.bold),
15          ), // Text
16          backgroundColor: Colors.blue,
17          foregroundColor: Colors.white,
18          centerTitle: true,
19          elevation: 0,
20        ), // AppBar
21        body: Obx(
22          () => Container(
23            decoration: BoxDecoration(
24              gradient: LinearGradient(
25                begin: Alignment.topCenter,
26                end: Alignment.bottomCenter,
27                colors: [Colors.blue.shade50, Colors.white],
28              ), // LinearGradient
29            ), // BoxDecoration
30            child: Padding(
31              padding: EdgeInsets.all(20),
32              child: Column(
33                crossAxisAlignment: CrossAxisAlignment.start,
34                children: [
35                  // Card info interval
36                  Card(
37                    elevation: 2,
```

Gambar 1. 12 source output_page.dart

```
main.dart M   data_diri_controller.dart ●   input_page.dart   output_page.dart ×   widget_test   BLACKBOX
lib > pages > output_page.dart > ...
5   class OutputPage extends StatelessWidget {      Info: Constructors for public widgets should ha
9     Widget build(BuildContext context) {
38                  shape: RoundedRectangleBorder(
39                    borderRadius: BorderRadius.circular(10),
40                  ), // RoundedRectangleBorder
41                  color: Colors.blue.shade50,
42                  child: Padding(
43                    padding: EdgeInsets.symmetric(horizontal: 15, vertical: 10),
44                    child: Row(
45                      children: [
46                        Icon(Icons.timer, color: Colors.blue, size: 20),
47                        SizedBox(width: 10),
48                        Expanded(
49                          child: Text(
50                            "Interval ngecek data setiap 1 detik",
51                            style: TextStyle(fontWeight: FontWeight.w500),
52                          ), // Text
53                        ), // Expanded
54                        Container(
55                          width: 8,
56                          height: 8,
57                          decoration: BoxDecoration(
58                            color: Colors.green,
59                            shape: BoxShape.circle,
60                          ), // BoxDecoration
61                        ), // Container
62                      ],
63                    ), // Row
64                  ), // Padding
65                ), // Card
66
67                SizedBox(height: 20),
68
69                // Notifikasi perubahan data (blink effect)
70                if (controller.isDataChanged.value)
71                  AnimatedContainer(
72                    duration: Duration(milliseconds: 300),
```

Gambar 1. 13 source output_page.dart

```
main.dart M   data_diri_controller.dart ●   input_page.dart   output_page.dart ×   widget_test   BLACKBOX
lib > pages > output_page.dart > ...
5   class OutputPage extends StatelessWidget {      Info: Constructors for public widgets should ha
9     Widget build(BuildContext context) {
73                    padding: EdgeInsets.all(12),
74                    decoration: BoxDecoration(
75                      color: Colors.green.withOpacity(0.2),      Info: 'withOpacity' is depre
76                      borderRadius: BorderRadius.circular(10),
77                      border: Border.all(color: Colors.green, width: 2),
78                    ), // BoxDecoration
79                    child: Row(
80                      children: [
81                        Icon(
82                          Icons.notifications_active,
83                          color: Colors.green,
84                          size: 24,
85                        ), // Icon
86                        SizedBox(width: 10),
87                        Expanded(
88                          child: Text(
89                            "🚀 ADA PERUBAHAN DATA! Interval mendeteksi perubahan!",
90                            style: TextStyle(
91                              color: Colors.green.shade800,
92                              fontWeight: FontWeight.bold,
93                            ), // TextStyle
94                          ), // Text
95                        ), // Expanded
96                      ],
97                    ), // Row
98                  ), // AnimatedContainer
99
100                 if (controller.isDataChanged.value) SizedBox(height: 20),
101
102                 // Title
103                 Text(
104                   "DATA TERKINI",
105                   style: TextStyle(
106                     fontSize: 18,
107                     fontWeight: FontWeight.bold,
```

Gambar 1. 14 source output_page.dart

```
main.dart M   data_diri_controller.dart ●   input_page.dart   output_page.dart ×   widget_test   BLACKBOX
lib > pages > output_page.dart > ...
5   class OutputPage extends StatelessWidget {      Info: Constructors for public widgets should ha
9     Widget build(BuildContext context) {
108                 color: Colors.blue.shade800,
109               ), // TextStyle
110             ), // Text
111             SizedBox(height: 5),
112             Text(
113               "Data yang berhasil disubmit akan muncul di sini",
114               style: TextStyle(color: Colors.grey[600], fontSize: 12),
115             ), // Text
116
117             Divider(height: 30),
118
119             // Display data
120             if (controller.submittedData.isEmpty)
121               Center(
122                 child: Column(
123                   children: [
124                     Icon(Icons.inbox, size: 80, color: Colors.grey[400]),
125                     SizedBox(height: 20),
126                     Text(
127                       "Belum ada data",
128                       style: TextStyle(
129                         fontSize: 18,
130                         color: Colors.grey[500],
131                       ), // TextStyle
132                     ), // Text
133                     SizedBox(height: 10),
134                     Text(
135                       "Submit data dulu di halaman input ya!",
136                       style: TextStyle(color: Colors.grey[400]),
137                     ), // Text
138                   ],
139                 ), // Column
140               ) // Center
141             else ...[
142               _buildInfoCard(
```

Gambar 1. 15 source output_page.dart

```
lib > pages > output_page.dart > ...
5   class OutputPage extends StatelessWidget {      Info: Constructors for public widgets should ha
9     Widget build(BuildContext context) {
142               _buildInfoCard(
143                 icon: Icons.person,
144                 label: "Nama Lengkap",
145                 value: controller.submittedData['nama'] ?? "-",
146                 color: Colors.blue,
147               ),
148               SizedBox(height: 12),
149               _buildInfoCard(
150                 icon: Icons.email,
151                 label: "Email",
152                 value: controller.submittedData['email'] ?? "-",
153                 color: Colors.orange,
154               ),
155               SizedBox(height: 12),
156               _buildInfoCard(
157                 icon: Icons.cake,
158                 label: "Umur",
159                 value: controller.submittedData['umur'] == "-"
160                     ? "Belum diisi"
161                     : "${controller.submittedData['umur']} tahun",
162                 color: Colors.purple,
163               ),
164               SizedBox(height: 12),
165               _buildInfoCard(
166                 icon: Icons.access_time,
167                 label: "Waktu Submit",
168                 value: controller.submittedData['waktu_submit'] ?? "-",
169                 color: Colors.grey,
170               ),
171             ],
172
173             Spacer(),
174
175             // Tombol balik
176             SizedBox(
```

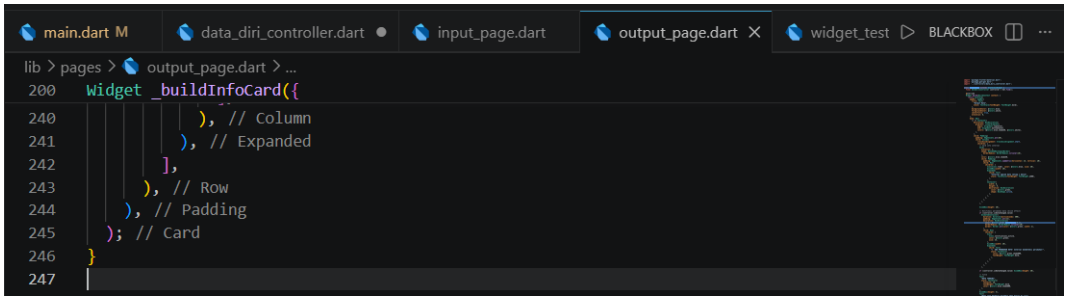
Gambar 1. 16 source output_page.dart

```
main.dart M   data_diri_controller.dart ●   input_page.dart   output_page.dart ×   widget_test   BLACKBOX
lib > pages > output_page.dart > ...
5   class OutputPage extends StatelessWidget {      Info: Constructors for public widgets should ha
9     Widget build(BuildContext context) {
175                 // Tombol balik
176                 SizedBox(
177                   width: double.infinity,
178                   child: OutlinedButton.icon(
179                     onPressed: () => Get.back(),
180                     icon: Icon(Icons.arrow_back),
181                     label: Text("KEMBALI KE INPUT"),
182                     style: OutlinedButton.styleFrom(
183                       padding: EdgeInsets.symmetric(vertical: 12),
184                       shape: RoundedRectangleBorder(
185                         borderRadius: BorderRadius.circular(10),
186                       ), // RoundedRectangleBorder
187                     ),
188                   ), // OutlinedButton.icon
189                 ), // SizedBox
190               ],
191             ), // Column
192           ), // Padding
193         ), // Container
194       ), // Obx
195     ); // Scaffold
196   }
197 }
198
199 // Widget helper buat info card
200 Widget _buildInfoCard({
201   required IconData icon,
202   required String label,
203   required String value,
204   required Color color,
205 }) {
206   return Card(
207     elevation: 2,
208     shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(12)),
209     child: Padding(
```

Gambar 1. 17 source output_page.dart

```
main.dart M   data_diri_controller.dart ●   input_page.dart   output_page.dart ×   widget_test   BLACKBOX
lib > pages > output_page.dart > ...
200 Widget _buildInfoCard({
206   return Card(
207     elevation: 2,
208     shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(12)),
209     child: Padding(
210       padding: EdgeInsets.all(15),
211       child: Row(
212         children: [
213           Container(
214             padding: EdgeInsets.all(10),
215             decoration: BoxDecoration(
216               color: color.withOpacity(0.1),      Info: 'withOpacity' is deprecated and shouldn
217               borderRadius: BorderRadius.circular(10),
218             ), // BoxDecoration
219             child: Icon(icon, color: color, size: 28),
220           ), // Container
221           SizedBox(width: 15),
222           Expanded(
223             child: Column(
224               crossAxisAlignment: CrossAxisAlignment.start,
225               children: [
226                 Text(
227                   label,
228                   style: TextStyle(color: Colors.grey[600], fontSize: 12),
229                 ), // Text
230                 SizedBox(height: 4),
231                 Text(
232                   value,
233                   style: TextStyle(
234                     fontSize: 16,
235                     fontWeight: FontWeight.bold,
236                     color: Colors.grey[800],
237                   ), // TextStyle
238                 ), // Text
239               ],
240             ), // Column
241           ), // Expanded
```

Gambar 1. 18 source output_page.dart



Gambar 1. 19 source output_page.dart